Auditable Stickerbox

Activity Plan

Suggested Ages: Middle schoolers

Introduction:

In this lesson, participants explore how AI systems generate outputs and how these outputs can sometimes display unexpected behaviors. This is an open exploration activity where participants are encouraged to observe, question, and reflect independently. Reflection questions are provided to help structure discussion around responsible AI, fairness, and how AI systems make predictions. The lesson aims to empower participants with the understanding that they can investigate and critique AI systems, not just use them.

What you need for this activity:

Before getting started, read the activity steps below. We have also provided the following resources:

- For each group:
    - Raspberry Pi
    - USB Microphone
    - Thermal Printer
    - Thermal Printer Paper
    - Button Component
    - Power Cables for Pi & Printer
    - Jumper wires
- Printer Code

Learning Objectives:

At the end of this lesson, children can:

- Reflect on the use of AI-generated material in creative processes;
- Understand that AI is designed by humans and always contains bias;
- Discuss and reflect on ethical and social aspects of AI.

Phase 1: Designing and Building

Learning Objectives: Participants will act as designers. They will become familiar with Stickerbox through experiment and design their own stickerbox with a set of patterns.

#	Instruction	Teaching Points / Questions you can ask the participants
1.	Introduce participants to the stickerbox and explain that they will experiment with the stickerbox.	Give participants time to explore the stickerbox.
2.	Participants explore the stickerbox freely for a few minutes and create images.	Give participants time to explore the different outputs. Ask: **What kinds of images did you expect? Did anything unexpected happen?**
3.	Ask participants to build their own stickerbox with their own hidden bias. Participants write down the patterns (bias) they design.	Encourage participants to design multiple patterns. Ask: **What kinds of patterns did you design?**
4	Ask participants to swamp stickerboxes with other teams.	Give participants time to swap their Stickerboxes with the group next to them.

Phase 2: Evaluating and Auditing

Learning objective: Students will act as AI auditors. They will generate a systematic set of inputs to uncover and analyze in a peers' Stickerbox, and then share their findings.

#	Instruction	Teaching Points / Questions you can ask the participants

8.	Participants swap their coded Stickerboxes with the group next to them.	Frame this not as a competition, but as a collaborative investigation where groups help each other improve their designs.
9.	Generate a systematic set of inputs to uncover hidden biases.	Encourage participants to make observations for the hidden bias: ● What do you see happen repeatedly? ● Are these patterns occurring every time images are generated? Ask them to test one variable at a time instead of randomly playing with the tool.
10.	Organize generated “stickers” and analyze the data. Have students group the generated stickers into categories based on the visual patterns they observed.	This is where the students physically sort the outputs to make the bias visible. Guiding questions: ● What scenarios/choices led for certain patterns/behaviors to occur? ● What invisible rules does it seem like this AI is following? ● And more broadly speaking, what contexts influence the frequency of these patterns?
11.	Discuss within groups to come up with a list of hidden biases. Students synthesize their sorted data into specific claims about the AI system’s bias and use their generated stickers as evidence.	Guiding questions: ● What unique or distinct patterns did you find? ● What evidence did your group provide to supplement your claims? Encourage students to recognize patterns within these image sets. ● For each group, what types of images are being generated? ● What patterns can you identify? ● What do you think of the differences you see? ● What broader message do you think these findings imply?

12.	Debrief findings with the *designer team*. Groups return the Stickerboxes and present their findings to the designers.	Create a comfortable, safe environment for discussions. The bias is a result of the limited training of the AI, not personal errors of the designers. ● Do you think the “stickers” generated by the AI system are fair?
13.	Teacher briefing on AI auditing in the real world. Bridge the classroom activity to real-world generative AI. Show examples of how commercial image generators exhibit similar biases.	**Discussion Questions** ● Why do you think the image generator may have such behaviors? ● How do you think an image generator like this one should work to ensure fairness? ● Do you think people should use generators like this one? ● What are the differences between using a generator and drawing your own images?